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**Vellore Institute of Technology**  
(Deemed to be University under section 3 of UGC Act, 1956)

**SCHOOL OF COMPUTER SCIENCE AND ENGINEERING**

**BCSE301P**

**SOFTWARE ENGINEERING LAB**

**ASSESSMENT 2**

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**Hostel Management System: An Efficient  
Solution for Streamlined Hostel  
Administration"**

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**1. Construct following Work Break Down Structure and Explain as per your project**

**SRS.**

- Process-Based Work Breakdown Structure
- Product-Based Work Breakdown Structure
- Role-Based Work Breakdown Structure
- Geography-Based Work Breakdown Structure
- Milestone and Deliverable-Based Breakdown Structure

**2. Construct ER Diagram of your Project.**

# HOSTEL MANAGEMENT SYSTEM

## 1) Process-Based Work Breakdown Structure

This structure organizes the project by phases of the development lifecycle, such as initiation, planning, execution, testing, and deployment. Each phase includes specific tasks like requirement gathering, risk analysis, coding, and system testing. It ensures a logical flow of activities and allows clear tracking of progress through sequential steps. This approach is ideal for projects with well-defined stages and milestones.

A process-based WBS focuses on the phases of the project lifecycle, breaking down tasks by processes involved in planning, executing, and delivering the project.

### 1. Initiation

- Identify project stakeholders and objectives.
- Gather initial requirements and define scope.
- Create and approve the Software Requirements Specification (SRS).

### 2. Planning

- Develop the project roadmap and timelines.
- Allocate resources and assign responsibilities.
- Identify risks and create mitigation plans.

### 3. Execution

- Develop the system modules (frontend and backend).
- Perform database integration and setup APIs.
- Conduct internal reviews of progress.

### 4. Testing

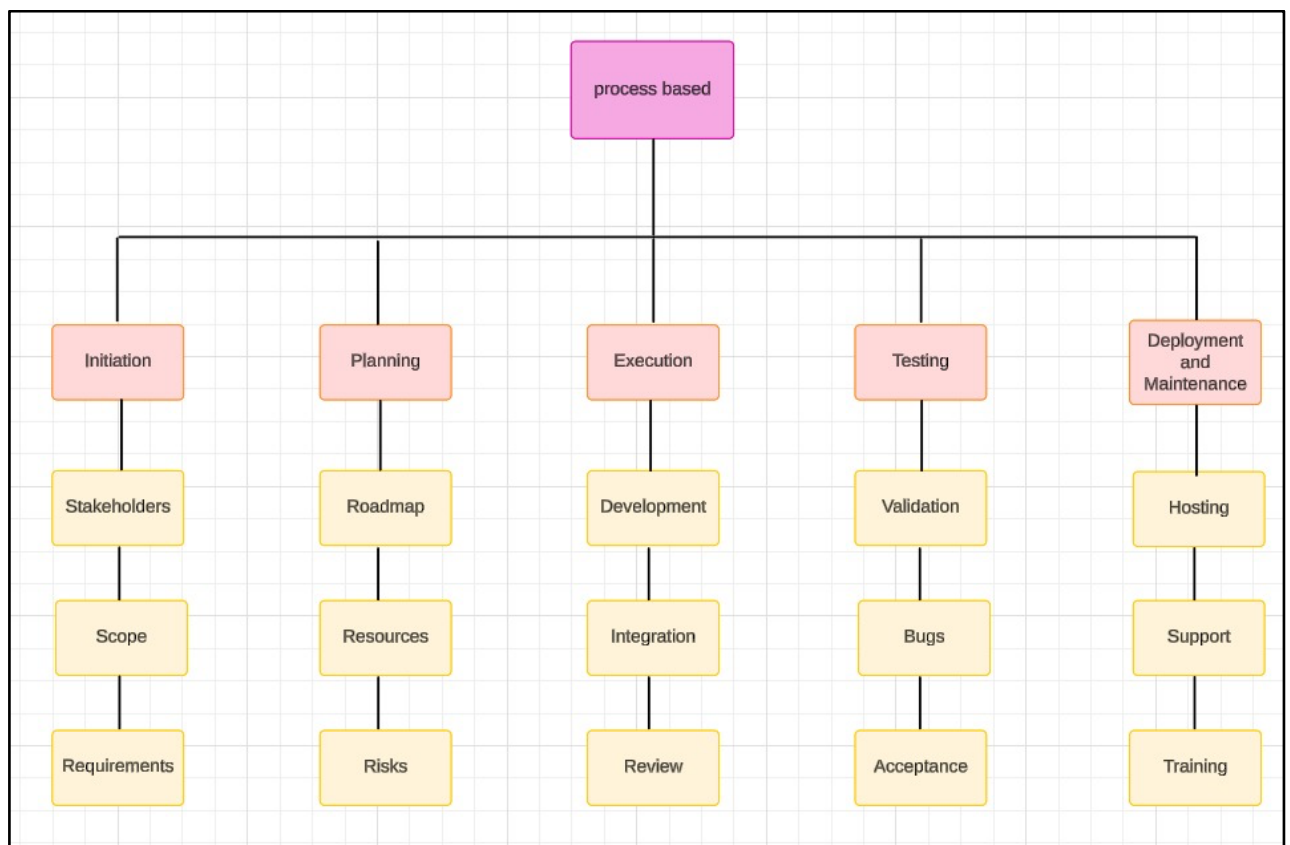
- Perform unit and integration testing.

- Conduct User Acceptance Testing (UAT) with stakeholders.
- Resolve identified bugs and retest.

## 5. Deployment and Maintenance

- Host the system on production servers.
- Provide user training and documentation.
- Offer ongoing support and feature updates.

## WBS – Work Breakdown Structure



## 2) Product-Based Work Breakdown Structure

Tasks are grouped based on the deliverables or modules of the project, such as authentication, room allocation, payment management, and grievance handling. Each module is further broken down into its components, such as frontend and backend development. This structure ensures a focus on achieving specific outputs or features. It is best suited for projects emphasizing modularity and feature development.

A product-based WBS organizes the tasks around deliverables or modules of the project.

### 1. User Authentication Module

- Design user registration and login interfaces.
- Implement secure authentication and password management.

### 2. Room Allocation Module

- Create a system for managing room bookings and availability.
- Generate reports for room allocation status.

### 3. Payment Management Module

- Develop payment tracking features.
- Integrate online payment systems.

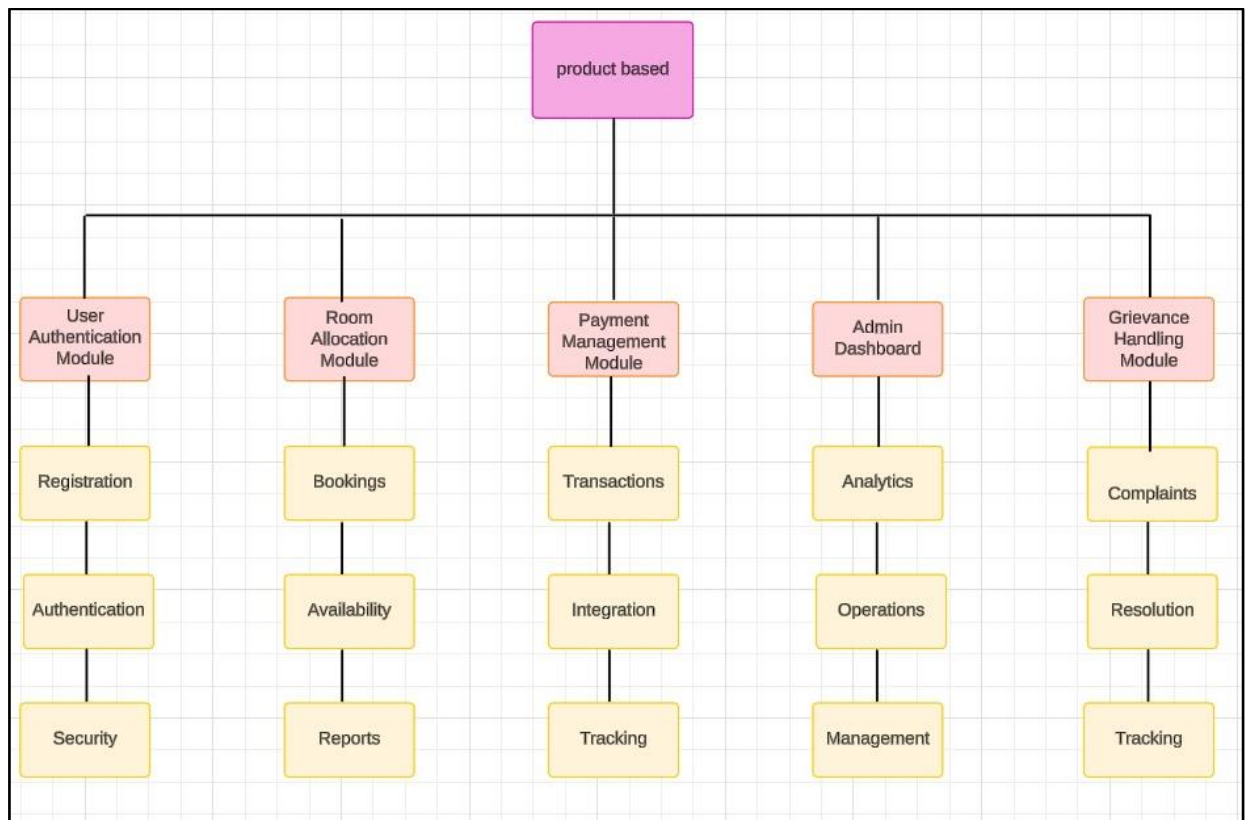
### 4. Admin Dashboard

- Build dashboards for hostel administrators to manage operations.
- Include analytics and reporting tools for decision-making.

### 5. Grievance Handling Module

- Allow users to log and track complaints.
- Enable admins to resolve and close grievances.

## WBS – Work Breakdown Structure



### 3) Role-Based Work Breakdown Structure

This structure categorizes tasks based on the roles or teams responsible, like project managers, developers, QA, and deployment teams. It clearly defines responsibilities, ensuring each team understands their deliverables. By mapping tasks to roles, it enhances collaboration and accountability among team members. It works well in projects with diverse teams handling distinct components.

A role-based WBS categorizes tasks based on the team or individual roles responsible for their completion.

#### 1. Project Manager

- Define project goals and objectives.
- Monitor progress and resolve issues.
- Communicate updates to stakeholders.

## 2. Frontend Development Team

- Design and implement user interfaces (HTML, CSS, JavaScript).
- Ensure responsive design across devices.

## 3. Backend Development Team

- Build database schemas and APIs.
- Ensure data security and system scalability.

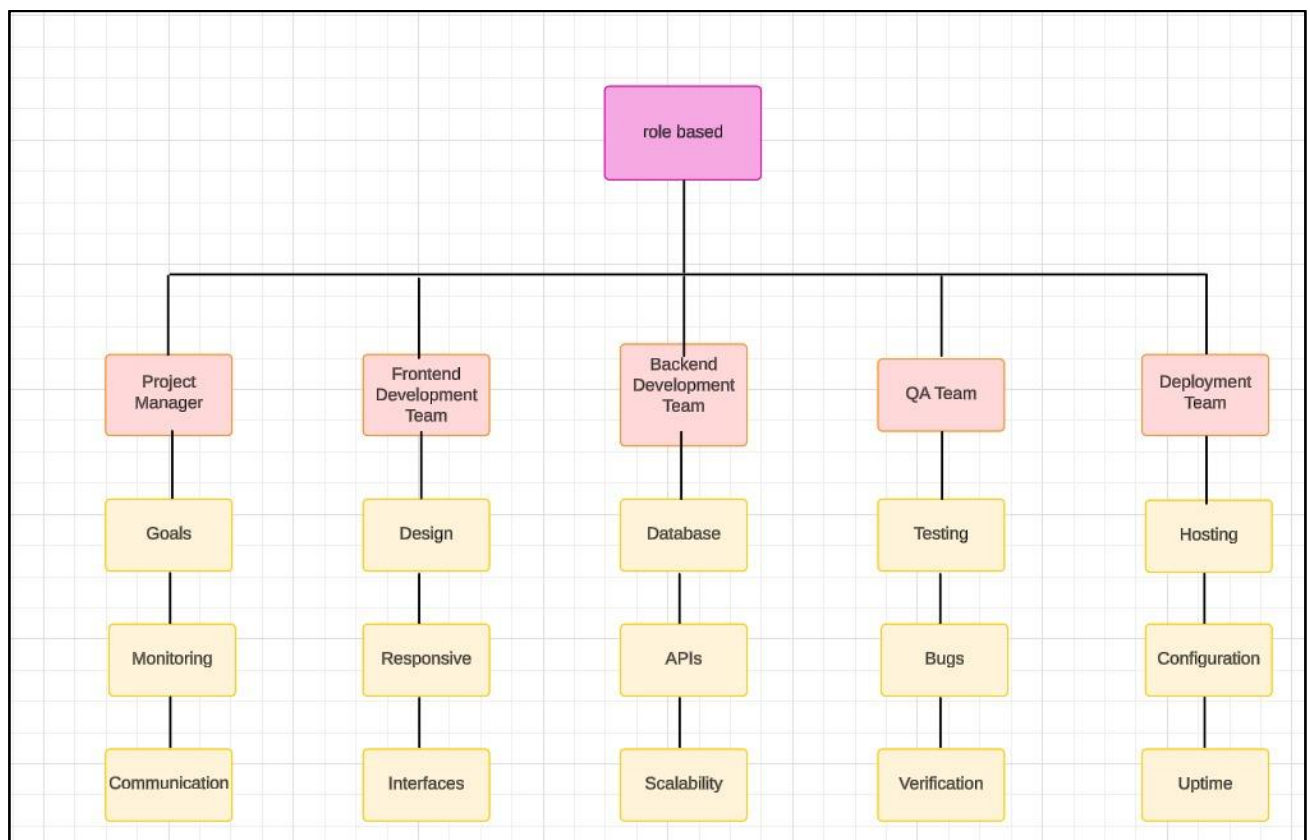
## 4. QA Team

- Create test cases and execute them.
- Document and track bugs, ensuring fixes are verified.

## 5. Deployment Team

- Configure and deploy the system to production.
- Set up hosting environments and ensure uptime.

### WBS – Work Breakdown Structure



#### **4) Geography-Based Work Breakdown Structure**

Tasks are divided based on the locations where the project is implemented, such as different hostels or campuses. Each location has tasks like requirement gathering, deployment, and feedback collection tailored to its specific needs. This approach ensures that regional requirements and constraints are considered. It is suitable for projects deployed across multiple geographical areas or remote environments.

A geography-based WBS is applicable when the project is implemented across multiple locations.

##### **1. Main Campus Hostel**

- Requirements gathering and stakeholder discussions.
- Deployment of modules specific to the main campus.

##### **2. Annex Hostel 1**

- Localized deployment and user training.
- Adapt features like room allocation to specific needs.

##### **3. Annex Hostel 2**

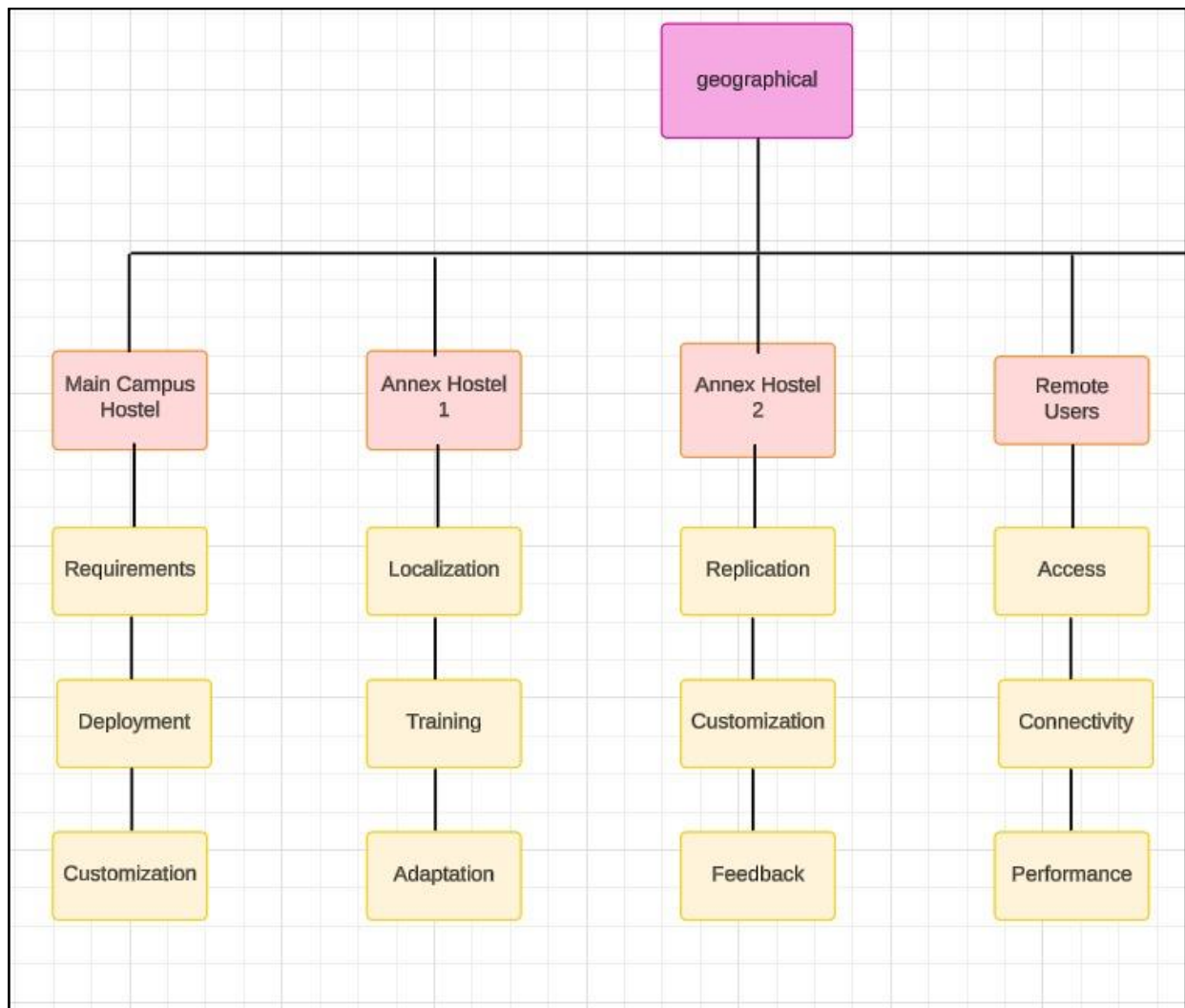
- Replicate and customize the system for unique use cases.
- Gather feedback to improve the central system.

##### **4. Remote Users**

- Set up online access for remote users.
- Address connectivity and performance challenges.



## WBS – Work Breakdown Structure



### 5) Milestone and Deliverable-Based Work Breakdown Structure

Tasks are organized around major milestones and their associated deliverables, like finalizing requirements, completing design, or successful testing. Each milestone marks a key checkpoint in project progress, ensuring timely completion of critical outputs. It provides a clear timeline and emphasizes the achievement of tangible results. This structure is ideal for tracking progress in deadline-driven projects.

Milestone 1: Requirements Finalized

- Complete the SRS document.

- Obtain approval from all stakeholders.

#### Milestone 2: Design Completed

- Deliver high-level and low-level design documents.
- Approve UI/UX prototypes.

#### Milestone 3: Development Completed

- Complete coding for all modules.
- Perform a successful internal review of functionality.

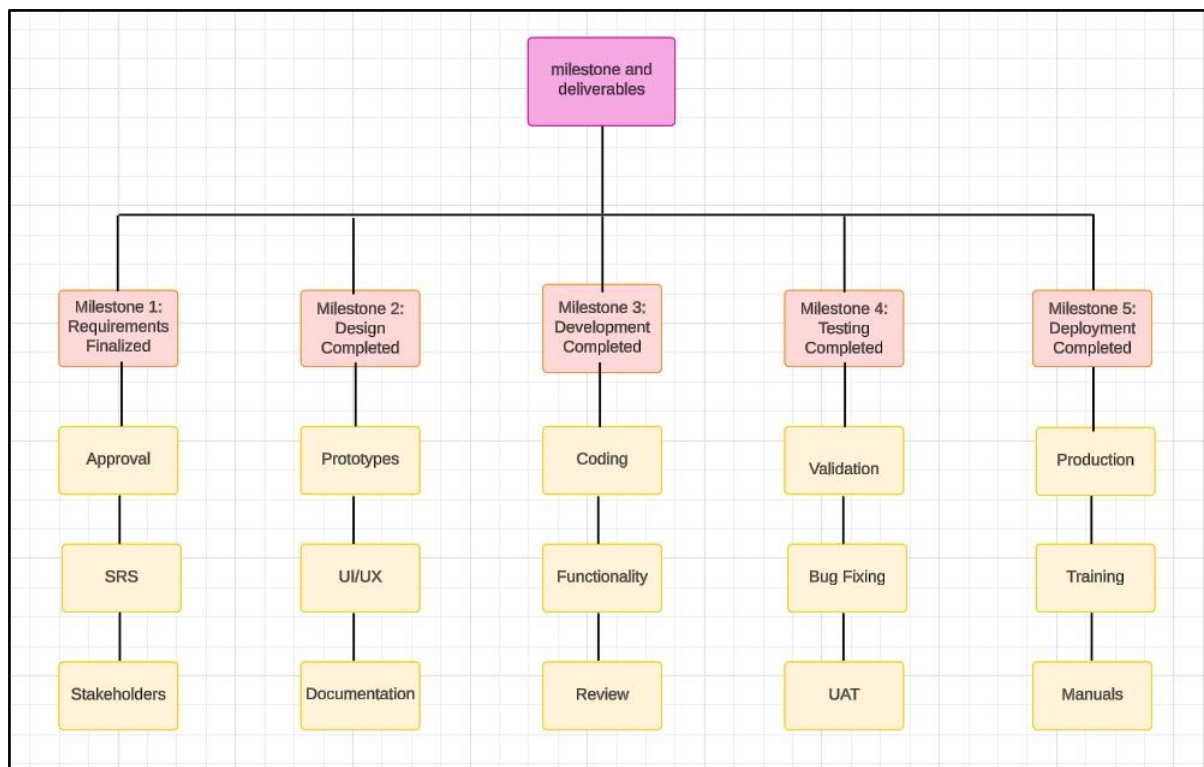
#### Milestone 4: Testing Completed

- Pass all test cases and resolve bugs.
- Obtain user approval after UAT.

#### Milestone 5: Deployment Completed

- Deploy the system in the production environment.
- Deliver user manuals and provide training.

### WBS – Work Breakdown Structure



## ER DIAGRAM

